
PROJECT REPORT: Windows Virtual Machine Deployment

Internship: Cloud AI with AWS

Organization: Cloudcredits Technologies Pvt. Ltd.

Task: Project No. 2

1. Project Objective

The objective of this project is to provision, configure, and remotely access a **Microsoft Windows Server** using **Amazon EC2**. This project demonstrates fundamental cloud infrastructure management, including instance launching, security group configuration, and RDP (Remote Desktop Protocol) connectivity.

2. Step-by-Step Implementation

Step 2.1: Provisioning the Windows Instance

1. Log in to the **AWS Management Console**.
2. Navigate to the **EC2 Dashboard** and click "**Launch Instance**".
3. **Name:** Set a name like `Windows_VM_Project`.
4. **AMI:** Select **Windows** and choose **Microsoft Windows Server 2022 Base** (Free Tier Eligible).
5. **Instance Type:** Select **t2.micro** to stay within the free tier.
6. **Key Pair:** Click "Create new key pair," name it `WindowsKey`, and download the `.pem` file.

Instance details | EC2 | ap-south-1

Console Home | Console Home

New Tab

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#InstanceDetails:instanceId=i-0681e27edd6538ad0

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EC2 > Instances > i-0681e27edd6538ad0

EC2

Dashboard

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Instances

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Snapshots

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Security Groups

Instance summary for i-0681e27edd6538ad0 (Windows-Server-Project2) Info

Connect Instance state Actions

Updated less than a minute ago

Instance ID

i-0681e27edd6538ad0

IPv6 address

-

Hostname type

IP name: ip-172-31-3-97.ap-south-1.compute.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

13.233.253.71 [Public IP]

IAM Role

-

IMDSv2

Required

Operator

-

Public IPv4 address

13.233.253.71 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-3-97.ap-south-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-00c5dc3ba1f98662c

Subnet ID

subnet-0362066909e59d222

Instance ARN

arn:aws:ec2:ap-south-1:961341508989:instance/i-0681e27edd6538ad0

Private IPv4 addresses

172.31.3.97

Public DNS

ec2-13-233-253-71.ap-south-1.compute.amazonaws.com | open address

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Managed

false

CloudShell Feedback Console Mobile App

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Step 2.2: Security and Network Configuration

1. Under **Network Settings**, ensure **RDP** traffic is allowed (Port 3389).
2. Set the source to **"My IP"** for secure access from your current location.
3. Click **"Launch Instance"**.

The screenshot displays the AWS Management Console for the 'ap-south-1' region. The main view is the 'Instances' page, showing a list of EC2 instances. One instance, 'i-0681e27edd6538ad0' (Windows-Server-Project2), is highlighted and its details are shown in the main panel. The instance is in the 'Running' state. The details panel includes sections for 'Instance summary', 'Public IPv4 address', 'Private IPv4 addresses', 'Public DNS', and 'Private IP DNS name'. The left sidebar contains navigation links for various AWS services, including EC2, S3, Lambda, IAM, CloudWatch, and CloudShell. The bottom of the console shows the Windows taskbar with various application icons and the system clock.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Publ
Windows-Serv...	i-0681e27edd6538ad0	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1b	ec2-

i-0681e27edd6538ad0 (Windows-Server-Project2)

Instance summary

Instance ID: i-0681e27edd6538ad0

IPv6 address: -

Hostname type: IP name: ip-172-31-3-97.ap-south-1.compute.internal

Public IPv4 address: 13.233.253.71 | open address

Instance state: Running

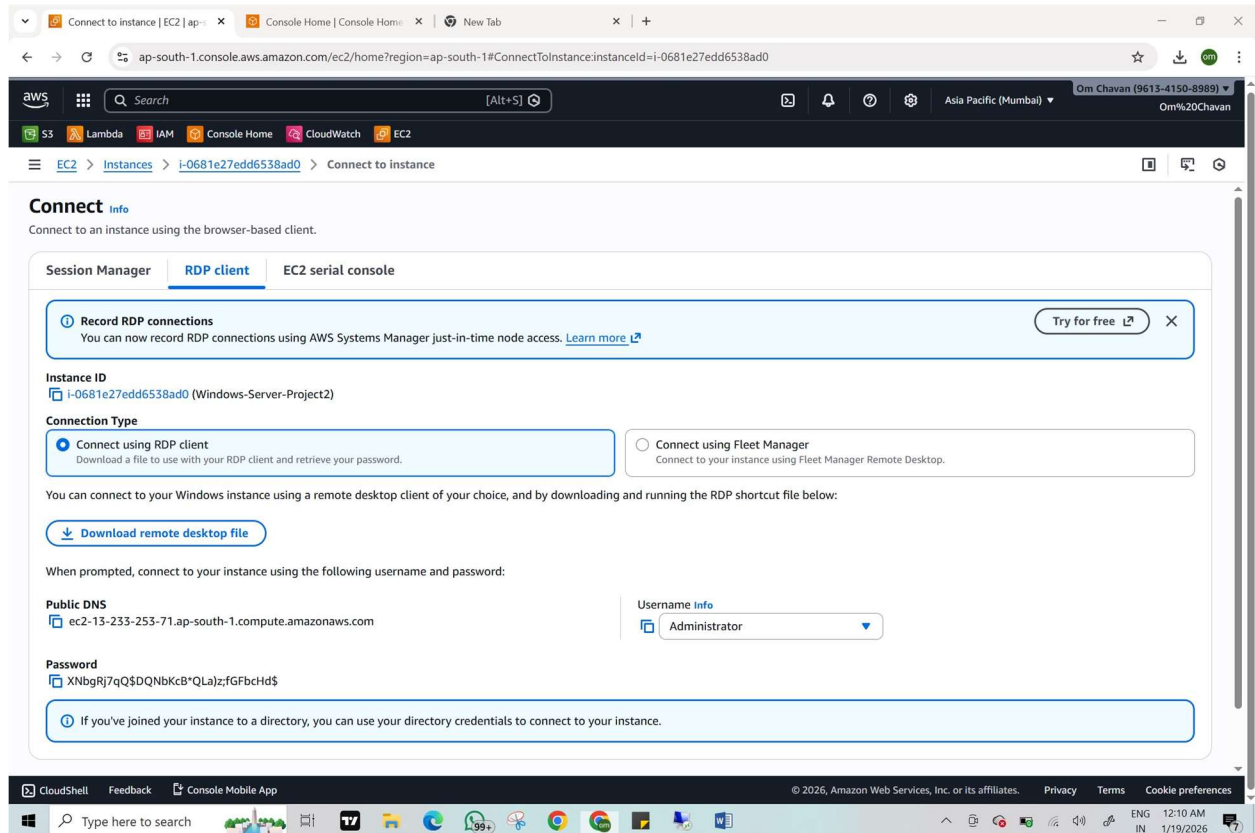
Private IP DNS name (IPv4 only): ip-172-31-3-97.ap-south-1.compute.internal

Private IPv4 addresses: 172.31.3.97

Public DNS: ec2-13-233-253-71.ap-south-1.compute.amazonaws.com | open address

Step 2.3: Password Decryption

1. Select your running instance and click **"Connect"**.
2. Navigate to the **RDP Client** tab and click **"Get password"**.
3. Upload the .pem key pair file downloaded in Step 2.1.
4. Click **"Decrypt Password"** and copy the resulting administrator password.



Step 2.4: Establishing Remote Desktop Connection

1. Open the **Remote Desktop Connection (RDP)** application on your local PC.
2. Enter the **Public IPv4 Address** of your instance.
3. Use the username `Administrator` and paste the decrypted password.
4. Accept the security certificate to log into the cloud desktop.

